EnergyLens

Power BI Report - Technical Documentation

Calculation methods, assumptions, data sources and energy logic for all 10 report pages

Energy Copilot Platform for Commercial Buildings

Microsoft Fabric - Medallion (Bronze/Silver/Gold) - DirectLake semantic model EnergyCopilotModel

Version: 2026-06-22

Contents

Right-click -> Update Field to build the table of contents.

1. About this document

This document explains what every page and every key visual of the EnergyLens Power BI report is built on: the calculation method, the assumptions and thresholds, the data source (gold table and measure), and the underlying energy logic. It is the single reference for the question 'what is this number, and how was it derived?'.

It serves two purposes: a refresher of the platform's own logic, and a decision-grade reference to show when the methodology is questioned.

How to read the source column

Each visual cites its gold table and the DAX measure name in square brackets, e.g. gold_kpi_daily - [Avg Solar PR Pct]. Measures live in the DirectLake semantic model EnergyCopilotModel; gold tables are produced by Fabric notebooks under the medallion architecture.

Data-basis legend (honesty layer)

Measured - computed from real telemetry/dispatch.

Modeled - derived from a physics or statistical model.

Prospect - a 'what-if' for an asset the building does not yet have.

Estimated (Est.) - a screening figure on a provisional baseline; disappears once metered data arrives.

Current data basis: *The platform today runs largely on synthetic data with realistic structure. Where a figure is real-telemetry-derived (e.g. the solar telemetry rollup, three measured battery dispatches) it is called out on the page. Section 15 details what is real vs synthetic.*

2. Architecture & data foundation

The platform follows a medallion architecture on Microsoft Fabric:

- Bronze - protocol-native raw ingestion (meters, weather, IoT/SCADA, solar, battery).
- Silver - cleaned, conformed, unit-normalised tables (e.g. silver_building_master, silver_weather_clean).
- Gold - business-logic tables the report reads (gold_kpi_daily/hourly/monthly, gold_anomaly_log, gold_recommendations, gold_ghg_scope, gold_hvac_analytics, gold_iot_*, gold_battery_*, gold_solar_daily, gold_consumption_forecast).
- Semantic model - a DirectLake model (EnergyCopilotModel) over the gold tables; ~213 measures; the 10 report pages bind to these measures.

Gold tables are mirrored to Supabase materialised views (mv_*) so the web application can read the same numbers without a live Fabric connection. Reads are pg-first with a Fabric fallback.

Refresh path: *A gold change requires re-running the notebook, refreshing the mv mirror (50_materialize), then refreshing the semantic model. A new gold column needs schema evolution (autoMerge) or an overwrite, since MERGE does not add columns.*

3. Global assumptions, thresholds & factors

Energy thresholds quoted across the report are screening-grade and were product-owner approved (2026-06-18). They are intended to triage attention, not to replace an audit.

Domain	Approved screening threshold / factor
EUI bands	GEG Anlage 10 (residential basis, approximate for commercial). Type benchmarks: Office 120-220, Hotel/Healthcare ~130, Logistics ~60 kWh/m2/yr; data centre not EUI-modelled.
Base-load ratio	<30% lean - 30-50% typical - >50% investigate.
Anomaly spike	demand > ~120% baseline / 2.5x rolling average; solar PR drop < 0.65.
Payback	<3 yr quick win - 3-7 yr standard - >7 yr strategic.
PV performance ratio	healthy 0.75-0.85 - <0.70 fault - >1.0 data error. DE specific yield ~900-1050 kWh/kWp/yr.
HVAC	heating delta-T 10-20 K - cooling delta-T 5-7 K - COP/JAZ ~3.0 floor.
Comfort / air	20-24 degC occupied - CO2 <800 good / 800-1500 fair / >1500 poor.
Grid emission factors	DE 0.363 location / 0.725 residual-market - TR ~0.442 kgCO2/kWh (year-indexed).
Electricity price	DE ~0.20-0.30 - TR ~0.14 EUR/kWh (country-indexed).
MEPS / EPBD	worst 15% / EPC G by 2030 - EPC F by 2033.
PV install ROI	capex 1050 EUR/kWp - DE feed-in ~0.075 EUR/kWh - fallback yield 950 kWh/kWp - NPV annuity factor 12 (~20 yr @ ~5%).
Battery	EU Battery Regulation 2023/1542; payback indicative, tariff- and self-consumption-dependent.

Page 1. Portfolio Overview

Module gate: meters (always on) **Data source:** *gold_kpi_daily, gold_kpi_monthly - DirectLake semantic model EnergyCopilotModel*

The whole portfolio at a glance - total energy, cost and CO2, and how buildings rank against one another. The first question it answers is where the savings sit.

Look at first: *The worst-ranked buildings on the EUI benchmark - that is where most of the savings are.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
EUI benchmark by building	Energy Use Intensity (annual kWh per m2) for each building vs the portfolio, so the least-efficient buildings stand out.	EUI = total_consumption_kWh / conditioned_area_m2, annualised. Area-weighted at portfolio level. Source: gold_kpi_daily - [P1 Building EUI], [P1 EUI Type Benchmark], [Avg EUI Area-Weighted].	GEG Anlage 10 bands (residential basis, approximate for commercial). Type benchmarks: Office 120-220, Hotel/Healthcare ~130, Logistics ~60 kWh/m2/yr; data centre not EUI-modelled. Bars to the right = least efficient.
Consumption trend (current vs prior year)	Year-over-year direction of total consumption; rising while floor area is flat signals creeping waste.	Period SUM with a prior-year comparison. Source: gold_kpi_daily - [Total Consumption kWh], [Previous Period Consumption kWh], [Consumption Change Pct].	Compared like-for-like (same calendar window). Direction matters more than the absolute in screening.
Cost & CO2 totals	The portfolio's financial and carbon exposure. CO2 drives CRREM/EPC risk even where the bill still looks acceptable.	Electricity cost = net_grid_kWh x country tariff; CO2 = net_grid_kWh x year-indexed grid factor. Source: gold_kpi_daily - [Total Energy Cost EUR], [Total CO2 Emissions tCO2].	KPI totals use a plain Card + SUM (not AVERAGE). EUI cards use AVERAGE deliberately - mixing the two is the classic card bug avoided here.
Building summary table	Per-building EUI, cost and EPC class - a triage list for where to spend attention first.	Row context per building over the selected period. Source: gold_kpi_daily + silver_building_master / silver_epc_ratings.	EPC class from area-weighted EPC score. Estimated rows are flagged where consumption is a provisional baseline, not metered.

Energy logic

- EUI is the primary efficiency screen: it normalises consumption by floor area so a small inefficient building and a large efficient one can be compared fairly.
- Portfolio carbon, not just cost, is surfaced because EU compliance (CRREM stranding, EPC/MEPS) is carbon-driven - a building can have a tolerable bill yet be a stranding risk.

Data basis: *Built on synthetic meter data. Ground-truth scale used for sanity checks: the data-centre building dominates at ~15M kWh/yr; portfolio ~32M kWh/yr. Where a building has no metered consumption, a provisional type-based baseline EUI range is shown and clearly marked Estimated.*

Page 2. Building Detail

Module gate: meters (always on) **Data source:** *gold_kpi_daily, gold_kpi_hourly*

One building's energy, cost and carbon over time, plus the core efficiency metrics that say whether it is run well.

Look at first: *Actual demand vs the baseline line - the gap above baseline is avoidable spend.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
EUI gauge	The headline 'is this building efficient?' number against typical bands.	EUI = consumption / conditioned area, annualised. Source: gold_kpi_daily - [P2 EUI Gauge ...], [Building EUI (kWh/m2, period)].	Fair/Poor = retrofit candidate. Commercial bands are approximate (GEG residential basis).
Consumption vs outdoor temperature	How weather-driven the building is. A steep slope means heating/cooling dominates; flat means plug/process loads.	Daily consumption plotted against avg outdoor temperature. Source: gold_kpi_daily - [Total Consumption kWh], [Avg Temperature C].	Steep slope -> envelope + HVAC levers. Flat -> equipment/process levers.
Daily load profile (24h)	When the building peaks, and the always-on overnight floor.	Average demand by hour of day. Source: gold_kpi_hourly - [Profile Avg Demand by Hour kW].	A high overnight floor is base load that scheduling can often trim without capex.
Base load ratio	Share of energy used when the building is effectively empty.	base (overnight/empty) demand / peak demand. Source: gold_kpi_hourly - [Base Load Ratio Pct].	Indicative: <30% lean - 30-50% typical - >50% investigate (higher is normal for refrigerated/logistics).
HDD / CDD vs consumption	Sensitivity to heating- and cooling-degree-days; tight cluster = predictable, outliers = possible anomalies.	Heating/cooling degree-days vs consumption. Source: gold_kpi_daily - [Total HDD], [Total CDD].	Degree-day base aligns with the climate-adjusted EUI normalisation used elsewhere.

Energy logic

- The baseline line is the expected demand; sustained operation above it is the avoidable component a facility manager can act on.
- Climate-adjusted EUI (ratio method on HDD+CDD) lets a mild year and a cold year be compared without crediting the weather.

Data basis: *Synthetic meter + weather data. Hourly profiles are modelled; when live IoT is connected (Page 8) the 24h profile can be driven by real readings.*

Page 3. Anomalies & Alerts

Module gate: meters (always on)   **Data source:** *gold_anomaly_log - rules in 03_gold_kpi_engine (section 8) + anomaly_detection.py*

Detected irregularities - spikes, drifts and faults that usually mean wasted energy - ranked so the costly ones surface first.

Look at first: *Critical and high anomalies first; each unresolved one is live, costed waste.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
Anomaly list by severity	Every detected event with severity, location and an estimated cost.	Counts and rollups over gold_anomaly_log. Source: [P3 Critical/High/Medium/Low Anomaly Count], [P3 Total Anomaly Count], [P3 Resolution Rate Pct].	Critical/high first. Severity is rule-specific (e.g. PR drop critical <0.50, high <0.60).
Detection rules (6)	The rule set that generates the log.	CONSUMPTION_SPIKE (demand > ~120% baseline / 2.5x rolling avg); NIGHT_OVERCONSUMPTION; SOLAR_PR_DROP (hourly PR < 0.65 while irradiance > 50 W/m2); BATTERY_OVERDISCHARGE (SoC < 10%); BATTERY_OVERCHARGE; DATA_QUALITY_GAP.	Thresholds are screening-grade and configurable (MIN_PR_THRESHOLD 0.65, spike 2.5x, SoC 10%).
Spike vs baseline / drift	One-off spikes (control fault, manual override) vs a slow upward creep (degrading equipment, loosened setpoints).	Event deviation vs rolling baseline. Source: gold_anomaly_log - [P3 Anomaly Deviation Pct].	Spike > ~120% baseline flags an event; sustained drift is the more expensive pattern.
Estimated EUR impact	A rough cost for each unresolved anomaly, used only to prioritise.	cost = duration_h x power_waste_kW x grid_price. Source: gold_anomaly_log cost columns.	Always shown as 'Est.'. Grid price DE ~0.20 / TR ~0.14 EUR/kWh by building country.

Energy logic

- The engine separates transient spikes from persistent drift on purpose: a spike is an event to investigate; a drift is a maintenance trend that compounds.
- SOLAR_PR_DROP is computed only above 50 W/m2 irradiance so that night/low-sun noise does not generate false faults.

Data basis: *Synthetic data with deliberately injected anomalies (e.g. a B001 August PR drop) used to verify each rule fires. Cost figures are estimates.*

Page 4. Forecast & Recommendations

Module gate: meters (always on) **Data source:** *gold_consumption_forecast (forecast)* + *gold_recommendations (measures)*

Where energy and cost are heading on current behaviour, plus the recommended measures to bend the curve - ranked by economic and compliance merit.

Look at first: *The measures ranked by saving vs payback - start at the top.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
Consumption / cost forecast	The 'do nothing' trajectory to beat, with a confidence band.	Pre-computed time-series forecast with prediction interval. Source: gold_consumption_forecast - [Forecast Predicted kWh], [Forecast Lower/Upper Bound kWh].	Forecast is precomputed in the forecast notebook; the measures simply read the predicted and bound columns.
Measures ranked by saving vs payback	Each retrofit/operational measure scored and ranked per building.	priority_score = 0.35 x compliance_urgency + 0.30 x financial(NPV/200k) + 0.20 x CO2_impact + 0.15 x payback_speed. Source: gold_recommendations - [Priority Sort Order], [Avg Payback Years].	Labels: CRITICAL >=80 - HIGH >=65 - MEDIUM >=45 - LOW >=25 - INFORMATIONAL <25. Payback: <3 quick - 3-7 standard - >7 strategic yr.
Economic viability gate (NPV)	An optional investment that loses money is not presented as an opportunity.	If npv_eur <= 0 and the action is optional, priority is capped to INFORMATIONAL. Applied AFTER the savings cap so it uses the final NPV. Source: gold_recommendations section 9c.	Compliance-mandated retrofits (heat pump, insulation, deep retrofit, audit) are EXEMPT - legal duty outranks ROI. With annuity factor 12, NPV<=0 is equivalent to payback >= 12 yr.
Saving potential per measure	Estimated annual EUR / kWh / CO2 for each measure.	Measure-specific sizing (e.g. INSTALL_SOLAR: roof area x yield x tariff; capex = kWp x 1050 EUR). Source: gold_recommendations.	Screening-grade - confirm with an audit before capex. A portfolio-realism cap scales a building's stacked savings to <=65% of its energy cost / 75% of emissions so totals stay physical.

Energy logic

- Scoring is deliberately compliance-weighted (0.55 of the score) because EU MEPS/EPBD are forcing functions independent of ROI - but the NPV gate stops a money-losing OPTIONAL measure from masquerading as an opportunity.
- The savings cap prevents the double-count that occurs when several measures each claim the same kWh off the same baseline.
- A capex realism floor keeps paybacks physical: engineered capex is sized per m2 while savings scale with kWh, so energy-dense buildings (data centre, hospital) produced impossible sub-year paybacks. Capex is floored to a realistic minimum implied by the building's own saving (no measure pays back faster than MIN_PLAUSIBLE_PAYBACK_YR, default 1.5 yr, once soft costs are included);

because the floor scales with the saving it only corrects the distorted high-intensity cases. Per-measure savings percentages were also separated so different measures stop returning identical euro figures.

Data basis: *Recommendation economics use real tariffs/feed-in and standard annuity assumptions; savings are screening estimates on synthetic baselines.*

Page 5. Occupancy Analysis

Module gate: meters (always on) **Data source:** *gold_occupancy_profile*

How occupancy drives energy use - separating base (empty) load from occupied load to expose the cheapest, no-capex savings.

Look at first: *High load when the building is empty - a scheduling / controls opportunity.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
Base (empty) vs occupied load	What runs when nobody is in the building.	Empty-hour demand vs occupied-hour demand. Source: gold_occupancy_profile - [Ghost Load Risk Pct], [Ghost Load Annual Cost EUR].	A high empty load (>~50% of peak) is the cheapest fix - scheduling, not capex.
Load vs occupancy hours	Whether consumption actually falls after hours.	Consumption aligned to occupancy schedule. Source: gold_occupancy_profile - occupancy-by-hour measures.	If load does not drop after hours, timers/controls are the lever.
Weekday / weekend pattern	Always-on equipment shows up as weekend use near weekday levels.	Avg occupancy by day-type. Source: [Avg Weekday/Weekend Occupancy Pct].	Confidence shown via [Avg Profile Confidence] / [Calibrated Buildings Count].

Energy logic

- Ghost load (energy with the building empty) is framed in EUR/yr because it converts a controls problem into a budget line a manager can prioritise.

Data basis: *Occupancy is modelled/calibrated from load shape and headcount inputs, not from turnstile data - treat as indicative.*

Page 6. Sustainability Compliance

Module gate: meters (always on) **Data source:** *gold_ghg_scope + gold_compliance_results*

The building's carbon footprint and where it stands against the EU compliance lines (CRREM, EPC/MEPS, CSRD/ESRS).

Look at first: *The Scope 1/2 split and the EPC / MEPS position.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
Scope 1 / 2 split	On-site combustion vs purchased electricity - the split says whether to electrify or green the supply.	Scope 1 = fuel x combustion factors; Scope 2 = electricity x grid factor. Source: gold_ghg_scope - [Scope 1 tCO2], [Scope 2 Location/Market tCO2e], [Scope 1 Share Pct].	Scope 2 method (location vs market) materially moves the number - disclosed explicitly.
Carbon intensity vs CRREM	kgCO2/m2/yr against the 2C decarbonisation pathway; the crossing year is the 'stranding' year.	Carbon intensity vs CRREM pathway. Source: [Carbon Intensity S1S2 kgCO2 m2 yr], [CRREM Pathway 2C], [CRREM Stranding Gap/Status].	Stranding = the year the building crosses the tightening pathway.
EPC / MEPS position	Where the building sits against EU renovation deadlines.	Area-weighted EPC. Source: [EPC Score/Class Area-Weighted].	MEPS: worst 15% / EPC G by 2030, EPC F by 2033. EPC F/G/H = renovation watch.
Grid factor used	Which emission factor underlies Scope 2.	Year-indexed factor from ref_grid_emission_factors. Source: gold_ghg_scope.	DE location-based ~0.363 vs residual-mix ~0.725 kgCO2/kWh; TR grid ~0.442. The choice moves Scope 2 a lot.

Energy logic

- Scope split is the decision tool: a Scope-1-heavy building wants electrification (heat pump); a Scope-2-heavy building wants a greener supply contract or on-site PV.
- Legal framing is deliberate: the platform reports 'ESRS-E1-aligned' figures, not 'CSRD-compliant' certification.

Data basis: *Gas/fuel data is not metered in the synthetic set; Scope 1 uses a proxy. A prior B009 carbon-levy artefact (~166k EUR synthetic) was corrected to the real GHG basis.*

Page 7. HVAC Analysis

Module gate: meters (always on)   **Data source:** *gold_hvac_analytics (mirrored to mv_hvac_analytics)*

Heating, cooling and ventilation performance - usually the single biggest consumer and the richest source of cheap savings.

Look at first: *Supply-return delta-T and runtime vs occupancy - over-conditioning is common.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
Supply-return delta-T	The temperature difference across the coil - a health check on heat transfer and pumping.	supply_temp - return_temp. Source: gold_hvac_analytics.	Heating design ~10-20 K, cooling ~5-7 K. A collapsed delta-T = low-delta-T syndrome (over-pumping, poor transfer).
Runtime vs occupancy	Conditioning that runs outside occupied hours.	HVAC runtime aligned to occupancy. Source: gold_hvac_analytics.	The most common and cheapest waste to cut.
Heat pump COP / JAZ	Coefficient of performance / seasonal performance factor.	Rated-COP based; degraded in cold = sizing or defrost issue. Source: gold_hvac_analytics - [HVAC Avg COP].	~3.0 is a floor; 3.5-4.5 air-source / 4-5 ground-source is healthy.
Setpoint deviation & insulation	Over-conditioning cost and envelope quality.	Each degC of over-conditioning ~ 2-5 kW extra. Insulation score recalibrated to a 0-anchor worst-stock baseline; heat-loss badge = transmission loss. Source: gold_hvac_analytics - [HVAC Ventilation Share Pct].	Small setpoint changes pay back fast. CO2 figures annualised (16.5k -> 7.1k correction).

Energy logic

- delta-T and runtime are first because they are free to fix (controls), before any talk of equipment capex.
- COP is anchored to rated performance so a healthy unit in a cold snap is not mis-flagged as failing.

Data basis: *Synthetic HVAC telemetry; COP/insulation recalibrated this finalisation pass for physical realism.*

Page 8. IoT Monitoring

Module gate: iot (gated)   **Data source:** *gold_iot_realtime + gold_iot_fdd - Event Hub -> EventStream -> KQL + Lakehouse*

Live sensor readings - real-time power, comfort and air quality - for facility and energy managers.

Look at first: *Zones outside the comfort band, and any sensors that have gone quiet.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
Live power (building + HVAC)	Real-time kW vs baseline, split building-total and HVAC-only.	Latest readings vs baseline, green/amber/red. Source: gold_iot_realtime.	A sustained red is money leaking right now.
CO2 by zone	Air-quality / ventilation adequacy per zone.	Latest CO2 ppm per zone. Source: gold_iot_realtime.	<800 ppm good - 800-1500 fair - >1500 poor (under-ventilation or over-occupancy).
Comfort band	Zones inside/outside the occupied comfort range.	Zone temperature vs setpoint band. Source: gold_iot_realtime.	~20-24 degC occupied. Outside = complaints and usually over-conditioning.
Sensor uptime + FDD cost	Which sensors are reporting, and the estimated cost of detected faults.	Fault detection & diagnosis with a cost estimate. Source: gold_iot_fdd - [IoT FDD Cost Today EUR], [priority_score]. Freshness = MAX(event_date).	Anomaly cost = duration x power_waste x grid price (HVAC 2-5 kW/degC, CO2 spike 1-3 kW; grid DE 0.20 / TR 0.14).

Energy logic

- Page 8 is isolated from Pages 1-7 by design (KQL hot path + Lakehouse), sharing only building_id, so the real-time stream never blocks the historical model.
- FDD cost is single-sourced from gold_iot_fdd so the alert count and the EUR figure can never disagree.

Data basis: *Synthetic sensor stream today; the SCADA edge-agent path (Phase A) lands real Modbus/telemetry into the same schema when a customer connects hardware.*

Page 9. Battery Strategy

Module gate: battery (gated) **Data source:** *gold_battery_simulation (nb 16, honest rebuild) + gold_battery_dispatch*

Battery dispatch economics - charge/discharge against tariff and solar - and whether a battery pays for a given building.

Look at first: *Payback and self-consumption - the sizing has to match the load profile.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
Charge/discharge vs tariff	Whether the battery hugs the price curve.	Dispatch vs day-ahead tariff and solar surplus. Source: gold_battery_dispatch - [V6 Charge/Discharge Rate], [V2 Avg Grid Price].	Charges off-peak / on solar surplus, discharges at peak = good dispatch.
Self-consumption	Share of solar kept on-site rather than exported.	self-consumed / generated with the battery. Source: gold_battery_simulation.	Most battery value comes from raising self-consumption, not arbitrage alone.
State-of-charge trend	Cycling health.	SoC over time. Source: gold_battery_dispatch - [V6 SoC Pct], [Avg Battery SoC Pct].	Healthy cycling between bounds; flat/idle = oversized or mis-scheduled.
Payback (by data basis)	Years to recover the install, by the building's installed strategy.	capex / annual_savings, installed strategy only. Source: gold_battery_simulation - [C2 Payback Years], data_basis column.	Tariff- and self-consumption-dependent - indicative, not a quote. EU Battery Regulation 2023/1542.

Energy logic

- The simulation labels every row Measured / Modeled / Prospect. Only three buildings have real dispatch (B001 self-consumption ~3.7k, B003 peak-shaving ~69.9k, B005 backup ~20.6k EUR/yr); the rest are physics models or batteryless prospects.
- Arbitrage is shown as an annual figure (~61k EUR), not a 3.3-year cumulative. Peak/ToU IRRs (40-64%) are optimistic but reproduced from the measured B003 case (~63%), not fabricated.

Data basis: *Three buildings = real measured dispatch; others modelled. Caps were removed in the honest rebuild so paybacks/NPV are faithful rather than flattered.*

Page 10. Solar Performance

Module gate: solar (gated) **Data source:** *gold_kpi_daily (synthetic KPI) + gold_solar_daily (telemetry rollup)*

On-site PV generation, self-consumption and export - and whether each system is performing to its physical potential.

Look at first: *Performance ratio and self-consumption - a low PR points to a fault; high export hints a battery could pay.*

Visual	What it shows	Calculation & data source	Thresholds & assumptions
KPI cards	Generation, self-consumption %, installed kWp, average PR, CO2 avoided, solar coverage %.	CO2 avoided = self_consumed_kWh x year grid factor (only self-consumed displaces grid). Coverage = generation / consumption. Source: gold_kpi_daily - [Total Solar Generated kWh], [Avg Solar PR Pct], [Solar Coverage Pct].	Self-consumption is worth more than export. Coverage is small for large buildings (e.g. hospital ~2%).
PV Performance Ratio (energy-basis)	Each system's PR, the core 'is this PV healthy?' metric.	Energy-basis PR = sum(gen) / (kWp x sum(reference-yield)), reference-yield = sum(hourly avg W/m2)/1000. Clamped at 0.95; portfolio is generation-weighted. Source: gold_kpi_daily - [Avg Solar PR Pct].	Healthy 0.75-0.85 - <0.70 fault (shading/soiling/inverter) - >1.0 = sensor/data error. Bars coloured red <70 / amber 70-75 / green >=75.
Self-consumption split	Solar self-consumed vs exported (rolling 12 months).	self vs export share of generation. Source: [Solar Self Consumed kWh Rolling 12m], [Solar Exported kWh Rolling 12m].	High export = a battery or load-shift could raise on-site value.
Specific yield by building	Annual kWh per kWp - orientation/shading/underperformance screen.	generation / installed kWp, annualised. Source: gold_kpi_daily - [Avg Solar Specific Yield kWh kWp].	DE typical ~900-1050 kWh/kWp/yr. Well below = orientation, shading or underperformance.
PV Underperformance - Est. Annual Loss	For each underperforming system, the generation and money left on the table.	lost_kWh = annual_gen x (0.80/PR - 1); valued at the building's own effective solar tariff (savings / self-consumed). Buildings at/above target drop out. Source: gold_kpi_daily - [PV Underperf Lost Gen kWh (Est)], [PV Underperf Loss EUR (Est)].	Target PR 0.80 (mid healthy band). Top case ~33k EUR/yr; portfolio ~53k EUR/yr recoverable. All 'Est.'.
Install-solar opportunities	Buildings without PV ranked as install candidates.	Sizing + ROI per candidate, subject to the NPV viability gate (Page 4). Source: gold_recommendations (INSTALL_SOLAR).	A negative-NPV candidate (e.g. low-tariff TR site) is labelled INFORMATIONAL, not a MEDIUM opportunity.

Energy logic

- PR was rebuilt to an energy basis this pass: the old daily PR averaged hourly ratios, which exploded on low-sun days (PR up to 5.6, impossible). Energy-basis PR is robust to that and is generation-weighted at the portfolio level.
- The honest consequence is that 5 of 8 systems now read below the 0.75 healthy line - previously hidden by the inflated average.

Data basis: *The KPI path is synthetic; the telemetry path (gold_solar_daily, real-ish rollup) independently computes the same energy-basis PR and lands at ~0.80. Residual systems sitting at the 0.95 clamp reflect synthetic generation calibrated slightly high - a generator-calibration item, separate from this report layer.*

Methodology notes (cross-cutting)

Performance Ratio - energy basis

PV PR is computed on an energy basis at daily and monthly grain: PR = sum(generation) / (installed kWp x sum(reference-yield)), where reference-yield for a day = sum(hourly average W/m2)/1000 (equivalent peak-sun-hours). This replaces an earlier average-of-hourly-ratios method that produced impossible PR>1 on low-sun days. PR is clamped at 0.95 and gated to days with meaningful sun (reference-yield > 0.5). The portfolio figure is generation-weighted, so a small or low-sun system cannot inflate it.

Recommendation scoring & the NPV viability gate

priority_score = 0.35 compliance-urgency + 0.30 financial(NPV/200k) + 0.20 CO2-impact + 0.15 payback-speed. The pipeline then applies three corrections in order: (1) a portfolio-realism savings cap scales a building's stacked savings to <=65% of its energy cost / 75% of emissions; (2) a capex realism floor sizes each operational measure's capex to a realistic minimum implied by its own saving (default >= 1.5 yr payback), so energy-dense buildings no longer show implausible sub-year paybacks, and per-measure savings percentages are separated so measures stop returning identical euro figures; (3) an economic-viability gate caps any OPTIONAL investment with NPV<=0 to INFORMATIONAL - no longer an opportunity. Compliance-mandated retrofits (heat pump, insulation, deep retrofit, audit) are exempt because a legal duty outranks ROI. The model's annuity factor is 12, so NPV<=0 is equivalent to payback >= ~12 years.

Anomaly rules

Six rules generate gold_anomaly_log: CONSUMPTION_SPIKE (>~120% baseline / 2.5x rolling), NIGHT_OVERCONSUMPTION, SOLAR_PR_DROP (hourly PR<0.65 above 50 W/m2), BATTERY_OVERDISCHARGE (SoC<10%), BATTERY_OVERCHARGE, DATA_QUALITY_GAP. Each carries an estimated EUR impact (duration x power-waste x grid price) for prioritisation only.

GHG & climate normalisation

Scope 2 uses a year-indexed grid factor (DE 0.363 location / 0.725 residual-market). Carbon intensity is tracked against the CRREM 2C pathway to derive a stranding year. Consumption is climate-normalised with a HDD+CDD ratio method so weather years are comparable.

Data provenance - synthetic vs real

Honesty is a design principle: the report never presents a modelled or synthetic number as if it were measured.

- Meter, weather, HVAC and occupancy data are synthetic with realistic structure; figures derived from them are screening-grade.
- Solar has two paths: the synthetic KPI path (gold_kpi_daily) and a real-ish telemetry rollup (gold_solar_daily) that independently computes the same energy-basis PR (~0.80). The web app shows a data-basis badge.
- Battery: only three buildings have real measured dispatch (B001/B003/B005); others are physics models (Modeled) or batteryless (Prospect), tagged in gold_battery_simulation.
- Recommendations use real tariffs, feed-in and grid factors, but savings are estimates on synthetic baselines.
- Every estimate is labelled 'Est.' and provisional baselines disappear once metered consumption is connected.

Known residuals: *A few solar systems sit at the PR clamp (0.95) because synthetic generation is calibrated slightly high for those buildings - a generator-calibration item tracked separately (the total-energy-ledger workstream), not a report-layer defect.*

Glossary

Term	Definition
EUI (Energy Use Intensity)	Annual energy per conditioned floor area (kWh/m2/yr). The primary cross-building efficiency yardstick.
PR (Performance Ratio)	Actual PV yield divided by the theoretical yield from the irradiance received. 0.75-0.85 healthy; >1.0 is a data error, not real over-performance.
Specific yield	PV generation per installed kWp (kWh/kWp/yr). Germany typical ~900-1050.
Self-consumption	Share of on-site generation used on-site rather than exported; usually more valuable than export.
COP / JAZ	Heat-pump coefficient of performance / seasonal performance factor. ~3.0 floor; higher is better.
HDD / CDD	Heating / cooling degree-days - a weather measure used to climate-normalise consumption.
Scope 1 / 2 / 3	GHG Protocol categories: on-site combustion / purchased electricity / value-chain emissions.
CRREM	Carbon Risk Real Estate Monitor decarbonisation pathway; the year a building crosses it is its 'stranding' year.
MEPS / EPBD	EU Minimum Energy Performance Standards under the Energy Performance of Buildings Directive: worst 15% / EPC G by 2030, EPC F by 2033.
NPV	Net Present Value - lifetime savings discounted to today minus upfront cost. NPV<=0 means the investment does not pay back in present-value terms.
Payback	Years for cumulative savings to equal the upfront cost (undiscounted). With the model's annuity factor 12, NPV<=0 corresponds to payback >= ~12 yr.
data_basis (Measured/Modeled/Prospect)	Battery/solar provenance tag: real dispatch vs physics model vs a site without the asset.